YASHAN LIN

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EDUCATION

Universiti Malaya
Master of Systems Engineering | GPA: 3.67/4.00 Mar 2025 – Present

Putian University
Bachelor of Engineering in Robotics Engineering | GPA: 86.6/100.0 (Rank 3/304) Sep 2020 – Jun 2024

RESEARCH INTERESTS

My research interests lie in reinforcement learning for robot locomotion, with a focus on enabling robots to acquire complex dynamic behaviours. I am broadly curious about embodied AI and actively exploring directions at the intersection of robot learning and physical systems.

ACADEMIC PAPERS

Y, Lin., & M, Dahari. (2026). Reinforcement Learning-Controlled Quadruped Robot: From Basic Gaits to Complex Motions. *The 2nd International Transdisciplinary Research Conference (INTRAC 2026)*, Universiti Malaya. (Under Review)

RESEARCH EXPERIENCE

Reinforcement Learning-Controlled Quadruped Robot: From Basic Gaits to Complex Motions
Universiti Malaya | Advisor: Prof. Mahidzal Dahari Nov 2025 – May 2026 *Kuala Lumpur, Malaysia*

- Developed a three-stage quadrupedal locomotion training framework for Unitree Go2 using PPO in Isaac Lab with 2,048 parallel environments.
- Trained and evaluated policies for trotting, four-legged synchronised jumping with approximately 10 cm jump height, and lateral alternating jumping with RSI and AMP, followed by sim-to-sim validation in MuJoCo.
- Designed stage-specific reward functions for gait synchronisation, foot clearance, flight-phase control, landing stability, and lateral alternation, combined with curriculum learning and domain randomisation.
- Implemented RSI with 8 manually designed keyframes and AMP refinement for Stage 3, with ablation results showing improved coordination and motion quality in lateral alternating jumping.

Velocity-Tracking Locomotion with Adversarial Motion Priors
Universiti Malaya | Advisor: Prof. Mahidzal Dahari Jun 2025 – Aug 2025 *Kuala Lumpur, Malaysia*

- Implemented Adversarial Motion Priors (AMP) for a velocity-tracking task on Unitree Go2 using motion reference data from the open-source `legged_control`.
- Evaluated the trained AMP policy through Isaac Lab rollout visualisation and MuJoCo sim-to-sim validation for quadrupedal motion imitation.
- Learned behaviour is fundamentally constrained by reference data coverage. A forward-only dataset failed to generalise to backward locomotion.

Design of a Multifunctional Foot Bath Robot Based on Microcontroller Nov 2023 – Apr 2024
Putian University | Advisor: Jialing Lian *Putian, China*

- Independently designed and implemented a full-stack embedded system based on STM32F103RCT6, integrating 8 subsystems including motion control, temperature regulation, UWB positioning, and remote APP control.
- Developed PID-based temperature control (± 0.5°C accuracy) with DS18B20 sensor and trapezoidal acceleration/deceleration algorithm for stepper motor motion control via PWM.
- Achieved 3 cm UWB positioning accuracy; designed mechanical structure independently using SolidWorks.

YOLOv8-Based Automated Waste Classification and Sorting System Oct 2022 – Mar 2023
Fujian Province Engineering Training & Innovation Competition, *First Prize (Special)*
Putian University | Advisor: Min Chen *Putian, China*

- Developed an automated waste sorting system using a dual-conveyor belt mechanism with YOLOv8-based real-time classification of 4 waste categories deployed on Raspberry Pi.
- Implemented STM32-controlled servo routing to direct classified waste into corresponding bins based on detection results.

Autonomous Navigation and Precision Shooting Robot Mar 2022 – Aug 2022
China Robot and Artificial Intelligence Competition, *National Second Prize*
Putian University | Advisor: Prof. Jianhong Fan *Putian, China*

- Developed autonomous navigation and target shooting across 5 waypoints (3 shooting positions + start/end) on a competition-provided platform using ROS1.
- Calibrated waypoint coordinates and actuation timing to achieve precise in-zone positioning and water balloon launching at each target.

AWARDS & HONORS

First-Class Excellence Scholarship, Putian University	2021, 2022, 2023, 2024
Merit Student, Putian University	2021, 2022, 2023, 2024
First Prize (Special), Fujian Province Engineering Training & Innovation Competition	2023
Second Prize, Fujian Province Lanqiao Cup Software & IT Talent Competition	2023
Second Prize, Fujian Province MCU Application Design Competition	2023
National Second Prize, China Robot and Artificial Intelligence Competition	2022

SKILLS

Programming: Python, C/C++, MATLAB, Linux, LaTeX
Frameworks & Tools: Isaac Lab, MuJoCo, PyTorch, ROS1, ROS2, Git, STM32, SolidWorks
Languages: English (PTE: 58), Mandarin Chinese (Native)